Requirements tracker

Every requirement, success metric and test case from the Phase A book, where it is tested in Phase B, and the result. Filled in sprint by sprint as each one is reviewed; an empty row has not been reviewed yet.

Status: Pass · Fail · Partial · Measured (reported against a baseline, no pass mark) · Not measured · Not started

Functional requirements (FR)

ID	What it says (Phase A)	Where it is tested	Result	Status
FR-1a	Accept video from live camera feeds and from uploaded files	Sprint 2: camera test, upload and probe. Sprint 4: live capture (OMNI-312, OMNI-313)	Uploads stored with correct metadata; live: about 4 frames/s stored, stops in 0.1 s, recovers by itself in 11–20 s after the source is killed	**Pass** (sprint 4)
FR-1b	Keep the most informative frames by detecting movement	Algorithm 1 on files (OMNI-304, OMNI-305) and live (OMNI-501); retention on 38 dataset clips (OMNI-318)	The gate separates still from moving footage; kept frames: fixed cameras 4.7–19.8%, daylight dash-cam 57%. Live: every frame checked (15/s), 20–22% kept on `cam05`, counters balance	**Pass** on fixed cameras, **weak** on moving cameras (sprint 5)
FR-2	Turn each accepted segment into a searchable representation of what it shows and how it changes	OMNI-704, 706, 708, 710	Every kept frame gets a scene vector (SigLIP 2, 1152) and an action vector (InternVideo2, 512, D-33); on the laptop cameras get the scene vector only (D-32)	**Pass** (sprint 7–8; cameras: scene only on the laptop)
FR-3	Store each analysed segment with its time, source camera and a representative image	`videos` table (sprint 2); `segments` table (OMNI-304, OMNI-316); Qdrant (OMNI-703, 710)	A row per frame with time, source, score and reason; image files equal the kept rows in every run. Sprint 7–8: both vectors in Qdrant with their details (TC-3)	**Pass** (sprint 3–4, vectors sprint 7–8)
FR-4	Search stored video in plain language, returning a ranked list of the most relevant moments	OMNI-707, 711, 712, 716	Top 5 moments with time, score and what matched; a click plays that second; Compare shows scene, action and both side by side. Search lives inside each video and the live viewer, on that footage only (D-39). Quality: SM-1	**Partial** (per video only, D-39)
FR-5	Predict likely upcoming events from recent activity: event type, time until it, confidence			

ID	What it says (Phase A)	Where it is tested	Result	Status
FR-6	Check each predicted event and raise an alert only if it is credible			
FR-7	Generate a structured incident report with the evidence, video segments and timeline			
FR-8	Let authorised external software use search, analysis, prediction and live alerts			
FR-9	Let users search videos, view live alerts and review incident reports	The camera wall (OMNI-503 to 508, 512); the Operations Dashboard and Datasets screen (OMNI-610 to 612)	Wall, tile states, enlarge, rewind, layouts pass. Dashboard counts equal a direct count, every figure links; summary in 41 ms (p50)	**Partial** (sprint 6: live monitoring and dashboard; search and alerts later)
FR-10	Authenticate users before they can use any feature	TC-10 (`services/query/tests/test_tc10_auth.py`); the sign-in page	Sign-in works; protected routes refuse anyone without a valid token (TC-10, 15 of 15)	**Pass** (sprint 1)

Non-functional requirements (NFR)

ID	What it says (Phase A)	Where it is tested	Result	Status
NFR-1	A captured frame is ready to search in under 5 s	First half, capture to stored (OMNI-316, OMNI-502); complete (OMNI-714)	Sprint 4: 49 ms alone, 6.14 s with a long file. Sprint 5: max 0.15–0.37 s even with a long upload. Sprint 7–8, capture to searchable, one camera, 10 min: max 3.35 s with the GPU's encode time (D-31); 270 s on the laptop's CPU	**Pass** (combined figure, D-31)
NFR-2	Search results appear in under 3 s	OMNI-714	1,800 searches through nginx (50 queries, 3 rounds, 3 modes): slowest 2.02 s on the laptop's CPU, 0.22 s with a GPU text encode	**Pass**
NFR-3	A forecast is ready in under 2 s			
NFR-4	Handle at least 10 live feeds at once without slowing down	OMNI-317 (preliminary); OMNI-509, OMNI-510a, the NFR-4 revision (D-19); OMNI-613	Sprint 4: 8 sustained. Sprint 5, every frame checked, real footage, wall open: 3 sustained for 10 min; 4 not; 10 not, on a 4-core laptop. Sprint 6: faster grid calculation, identical results, still 3	**Fail** against Phase A (10); **Pass** against the revised target (3 per machine of this class, D-19)

ID	What it says (Phase A)	Where it is tested	Result	Status
NFR-5	Data lost when any part fails or crashes = 0	Sprint 2: uploads. Sprint 3–4: OMNI-311, OMNI-319. Sprint 6: OMNI-601 (live, rechecked)	Uploads: 3 of 3. Frames: consumer killed, broker rebuilt (1,800 of 1,800 kept), ingestion restarted and killed: 0 lost. Rechecked for live on the sprint 5 pipeline: 0 lost, 32 of 32 checks, 2 runs	**Pass** in the cases tested (sprint 6)
NFR-6a	Each major part can be updated or replaced on its own			
NFR-6b	New prediction types can be added without changing existing parts			
NFR-7	100% of API endpoints and MCP tools reject requests without a valid token or role	TC-10 suite; role checks on the camera and video routes	All 4 protected routes refused a missing, expired or tampered token (401); the one admin-only route (`/admin/ping`) refused the wrong role (403). Sprint 2: camera writes are admin-only, the operator gets 403 (demo step 5)	**Pass** (sprints 1–2: the routes that existed then)
NFR-8	100% of traffic encrypted with TLS 1.2+; plaintext bytes = 0	Built: nginx with TLS 1.2 and 1.3 only, plain HTTP redirects to HTTPS	Not recorded in `results.md`	**Not measured**
NFR-9	Tokens expire after at most 15 min idle; account locks after 5 failed logins	`services/query/tests/test_nfr9_lockout_expiry.py`	2 of 2: 5 wrong passwords lock the account (the correct password is then refused too); an expired token is refused	**Pass** (sprint 1)

Success metrics (SM)

ID	What it says (Phase A)	Where it is tested	Result	Status
SM-1	Search quality: the right clip in the top 5 for at least 85% of queries, with a lift on action queries over frames only (VIRAT)	OMNI-713, Colab, 50 queries over 35,704 VIRAT frames	76.0% fused (60.0% with Algorithm 3 as written, 74.0% frame only, chance 25.1%); places 92%, actions 60% against 60%, no lift. 85% not met; no published evidence supports it (D-41)	**Measured** (D-41)
SM-2	Prediction accuracy: the real event in the top 3 predictions in at least 70% of cases, per domain			

ID	What it says (Phase A)	Where it is tested	Result	Status
SM-3	Early warning: warnings at least 15 s before the event (Phase A: on DoTA)	Revised (D-2): mean time-to-accident on CCD; target rule D-23 (OMNI-606)	Target **≥ 1.80 s**: half the held-out median pre-crash footage (3.60 s); written before measuring. No model yet	**Target set, not measured** (sprint 6)
SM-4	False alarms: at most 20% of warnings false, per domain (UCF-Crime)			
SM-5	Cross-domain learning: pre-training on two domains lifts the third by at least 10%			
SM-6	Forensic reports: a structured report for at least 2 multi-camera scenarios (AI City)			

Test cases (TC)

ID	What it checks (Phase A)	Where it is tested	Result	Status
TC-1	A live feed and an uploaded file at once; only frames with motion forwarded (FR-1a, FR-1b)	OMNI-316 (first half); OMNI-511 (complete, on the wall)	2 runs, 15 of 15 checks each: 0 decode errors, 0 violations, the camera Live throughout	**Pass, complete** (sprint 5)
TC-2	A frame and a clip embedding for each admitted segment (FR-2)	OMNI-704, 710	Frame: 136 of 136 (upload), 41 of 41 (camera). Clip: 34 of 34 (upload); cameras get none on the laptop (D-32)	**Pass** (cameras: frame only on the laptop)
TC-3	Each embedding stored in Qdrant with its metadata intact (FR-3)	OMNI-703, 710	1,000 of 1,000 in each collection, by id and by filtered search, every field equal, again after a Qdrant restart	**Pass**
TC-4	Plain-language search over at least 100,000 embeddings; right clip in the top 5 for 85% (FR-4)	OMNI-715	Fused recall@5 74% on 35,704 frames (D-37), frame only 72%; search at most 0.2 s, one cold first query 10.9 s (K-26). 85% not met; no published evidence supports it (D-41)	**Measured** (D-41)
TC-5	Predict early and accurately, bounded false alarms, transfer to an unseen domain (FR-5)			
TC-6	Four-tier verification suppresses benign clips and lowers false positives (FR-6)			
TC-7	An evidence-grounded report from multiple cameras after an operator confirms (FR-7)			

ID	What it checks (Phase A)	Where it is tested	Result	Status
TC-8	All capabilities reachable by external AI agents through MCP, including live alerts (FR-8)			
TC-9	The search, alerts and report views render without errors (FR-9)			
TC-10	Every endpoint protected by authentication and roles: right role 200, wrong role 403, no or bad token 401 (FR-10)	`services/query/tests/test_tc10_auth.py`	15 of 15 on the 4 protected routes (`/auth/me` , `/nav` , `/auth/logout` , `/admin/ping`); MCP tools do not exist yet	**Pass** (sprint 1)